

Spring Data JPA - Quick Example

 **spring** initializr

Project

☐ Gradle - Groovy
☐ Gradle - Kotlin
☒ Maven

Language

☒ Java
☐ Kotlin
☐ Groovy

Spring Boot

☐ 4.1.1 (SNAPSHOT)
☒ 4.1.0
☐ 4.0.8 (SNAPSHOT)
☐ 4.0.7
☐ 3.5.16

Project Metadata

Group

com.example

Artifact

orm-learn

Package name

com.example.orm-learn

Packaging

☒ Jar
☐ War

Configuration

☒ Properties
☐ YAML

Dependencies

ADD DEPENDENCIES... CTRL + B

Spring Boot DevTools

DEVELOPER TOOLS

Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

Spring Data JPA

SQL

Persist data in SQL stores with Java Persistence API using Spring Data and Hibernate.

MySQL Driver

SQL

MySQL JDBC driver.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

airline
book_store
bookstore
coll2
college
e4
e5
emp
emp2
emp3
emp4
emp_management
hosp
inventory_db
lib
new
new2
online2
ormlearn
Tables
Views
Stored Procedures
Functions
part
retail_store
schema
Administration

Query 1

Limit to 1000 rows

7 name VARCHAR(50)
8 }
9 }
10 INSERT INTO country VALUES ('IN','India');
11
12 INSERT INTO country VALUES ('US','United States of America');
13 SELECT * FROM country;

Result Grid

code name
IN India
US United States of America

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Schema: ormlearn

#	Time	Action	Message	Duration / Fetch
1	20:03:00	CREATE DATABASE ormlearn	1 row(s) affected	0.031 sec
2	20:04:36	USE ormlearn	0 row(s) affected	0.015 sec
3	20:05:25	CREATE TABLE country (code VARCHAR(2), PRIMARY KEY, name VARCHAR(50))	0 row(s) affected	0.062 sec
4	20:06:08	INSERT INTO country VALUES ('IN','India')	1 row(s) affected	0.015 sec
5	20:06:08	INSERT INTO country VALUES ('US','United States of America')	1 row(s) affected	0.000 sec
6	20:06:28	SELECT * FROM country LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000

File Edit Selection View Go Run Terminal Help

Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More

EXPLORER

ORM-LEARN
> .mvn
> src
> main
> java\com\example\orm_learn
> model
> Country.java
> repository
> CountryRepository.java
> OrmLearnApplication.java
> resources
> application.properties
> test
> .gitattributes
> .gitignore
> HELP.md
> mvnw
> mvnw.cmd
> pom.xml

application.properties

src > main > resources > application.properties

1 #
2 # Logging Configuration
3 # =====
4
5 logging.level.org.springframework=INFO
6 logging.level.com.cognizant=DEBUG
7
8 # Hibernate SQL Logs
9 logging.level.org.hibernate.sql=DEBUG
10 logging.level.org.hibernate.orm.jdbc.bind=TRACE
11
12 # =====
13 # MySQL Database
14 # =====
15
16 spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
17 spring.datasource.username=root
18 spring.datasource.password=Satara@123
19 spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
20
21 # =====
22 # JPA Configuration
23 # =====
24
25 spring.jpa.hibernate.ddl-auto=validate
26 spring.jpa.show-sql=true
27 spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect

EXPLORER

- ORM-LEARN
 - .mvn
 - src
 - main
 - java\com\example\orm_learn
 - model
 - Country.java

application.properties

Country.java

```
1 package com.example.orm_learn.model;
2
3 import jakarta.persistence.Column;
4 import jakarta.persistence.Entity;
5 import jakarta.persistence.Id;
6 import jakarta.persistence.Table;
7
8 @Entity
9 @Table(name = "country")
10 public class Country {
11
12     @Id
13     @Column(name = "code")
14     private String code;
15
16     @Column(name = "name")
17     private String name;
18
19     public Country() {
20     }
21
22     public String getCode() {
23         return code;
24     }
25
26     public void setCode(String code) {
27         this.code = code;
28     }
29
30     public String getName() {
31         return name;
32     }
33
34     public void setName(String name) {
35         this.name = name;
36     }
37 }
```

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EXPLORER

- ORM-LEARN
 - .mvn
 - src
 - main
 - java\com\example\orm_learn
 - model
 - Country.java
 - repository
 - CountryRepository.java

application.properties

CountryRepository.java

```
1 package com.example.orm_learn.repository;
2
3 import org.springframework.data.jpa.repository.JpaRepository;
4 import org.springframework.stereotype.Repository;
5
6 import com.example.orm_learn.model.Country;
7
8 @Repository
9 public interface CountryRepository extends JpaRepository<Country, String> {
10
11 }
```

File Edit Selection View Go Run Terminal Help

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EXPLORER

- ORM-LEARN
 - .mvn
 - src
 - main
 - java\com\example\orm_learn
 - model
 - Country.java
 - repository
 - CountryRepository.java
 - service
 - CountryService.java

application.properties

CountryService.java

```
1 package com.example.orm_learn.service;
2
3 import java.util.List;
4
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.Service;
7
8 import com.example.orm_learn.model.Country;
9 import com.example.orm_learn.repository.CountryRepository;
10
11 import jakarta.transaction.Transactional;
12
13 @Service
14 public class CountryService {
15
16     @Autowired
17     private CountryRepository countryRepository;
18
19     @Transactional
20     public List<Country> getAllCountries() {
21
22         return countryRepository.findAll();
23     }
24
25 }
26
```

Tip: Try the before imp

16 public class OrmLearnApplication {

OUTPUT

DEBUG CONSOLE

TERMINAL 3

PORTS

PROBLEMS

Filter (e.g. text, **/*ts, !**/n...

▼ pom.xml 1

⚠ (BOOT_VERSION_VALIDA

▼ CountryRepository.java src\ma

⚠ vscode-spring-boot(JAVA

▼ CountryService.java src\main\ja

⚠ vscode-spring-boot(JAVA

TERMINAL

powershell ⚠ + ▾

at java.base/java.net.SocksSocketImpl.connect(SocksSocketImpl.java:327) ~[na:na]
at java.base/java.net.Socket.connect(Socket.java:633) ~[na:na]
at com.mysql.cj.protocol.StandardSocketFactory.connect(StandardSocketFactory.java:144) ~[mysql-conn
-9.3.0.jar:9.3.0]
at com.mysql.cj.protocol.a.NativeSocketConnection.connect(NativeSocketConnection.java:53) ~[mysql-c
r-j-9.3.0.jar:9.3.0]
... 60 common frames omitted

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 23.331 s
[INFO] Finished at: 2026-06-27T23:26:43+05:30
[INFO] -----
PS C:\Users\chava\Downloads\orm-learn\orm-learn>